N^{th} root of a complex number

Key Points

To work out the n^{th} root of a complex number first make sure its in exponential form

$$z^n = ae^{bi}$$

Then the solutions are

$$z_k = \sqrt[n]{a}e^{\frac{(b+2\pi k)i}{n}} \text{ for } k = 0,1,..,n-1$$

Remember to adjust the argument so that it is within the range of $-\pi$ to π

The solutions to $z^n = 1$ are called the roots of unity

Basic Skills

Q1. For the complex number $z = 2(1+i)$

a) Show $z = 2\sqrt{2}e^{\frac{\pi}{4}i}$

b) Find z^2 in exponential form

c) Show that $z^6 = 512e^{-\frac{\pi}{2}i}$

d) Find a and b such that $z^6 = a + bi$

Q2. For the complex numbers $z = 2e^{\frac{\pi}{3}i}$, $w = 2e^{-\frac{2\pi}{3}i}$

a) Find z^2

b) Find w^2

c) Hence state two solutions to $v^2 = 4e^{\frac{2}{3}\pi i}$

Competency Skills

Q3. Solve the following complex root equations

a) $z^4 = 81e^{\frac{\pi}{8}i}$

b) $z^3 = 20e^{i}$

c) $z^5 = 7 - 8i$

Q4. Find all the complex numbers that solve $\sqrt[3]{i-1}$ in exponential form

Q5. **a)** Solve the equation $z^5 = 1$ for solutions z_k for $k = 0,1,2,3,4$

b) Plot the solutions, z_k for $k = 0,1,2,3,4$, on an argand diagram labelling them as A,B,C,D and E respectively

c) What shape does ABCDE form in the complex plane

Advanced Skills

Q6. Find all the solutions to the equation $(2z+3)^3 = 3 + 4i$

Q7. **a)** Show that $w = e^{\frac{2\pi}{5}i}$ is a fifth root of unity

b) Hence show that $1, w^2, w^3, w^4$ are the other fifth roots of unity

c) Show that $1 + w + w^2 + w^3 + w^4 = 0$

d) Hence simplify and find the value of $w^2 + w^5 + w^8 + w^{11} + w^{-1}$

Q8. **a)** Plot the three roots to the equation, $z^3 = 2 + 2i$, on an argand diagram labelling them A,B and C

b) Find the area of the triangle ABC and verify your answer by checking it less than the are of its enscribed circle

Extension

Q9. Following on from Q8) a point P lies on the circle through A,B and C using w, α, β, γ to represent their complex numbers respectively show that $|(w-\alpha)^2 + (w-\beta)^2 + (w-\gamma)^2| = 6$

Hint: Use exponential form for the complex numbers